ME 656: Autonomous Navigation for Mobile Robots

Simulation Lab 2 Report

Course:	ME 656 - Autonomous Navigation for Mobile Robots
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Assignment:	SimLab 2
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1 Overview

This report summarizes the path-planning and localization-uncertainty results for Simulation Lab 2. The robot was planned through the provided 2D obstacle workspace using a Rapidly-exploring Random Tree (RRT). For each generated path, a linear Kalman filter covariance propagation was used to estimate the expected localization uncertainty along the path.

The final run used 1000 RRT trials. From those trials, four representative paths were selected: the shortest path, the minimum terminal uncertainty path, the maximum terminal uncertainty path, and a best-tradeoff path. The best-tradeoff path was selected using a normalized cost function with equal weighting between path length and terminal uncertainty.

2 Selected Paths

Path type	Trial	Path length (m)	Terminal uncertainty, $\mathrm{tr}(P)$
Shortest path	193	100.00	47.42
Minimum uncertainty	442	108.00	9.56
Maximum uncertainty	752	130.00	251.39
Best tradeoff	811	103.10	13.55

Table 1: Selected path outcomes from 1000 RRT trials.

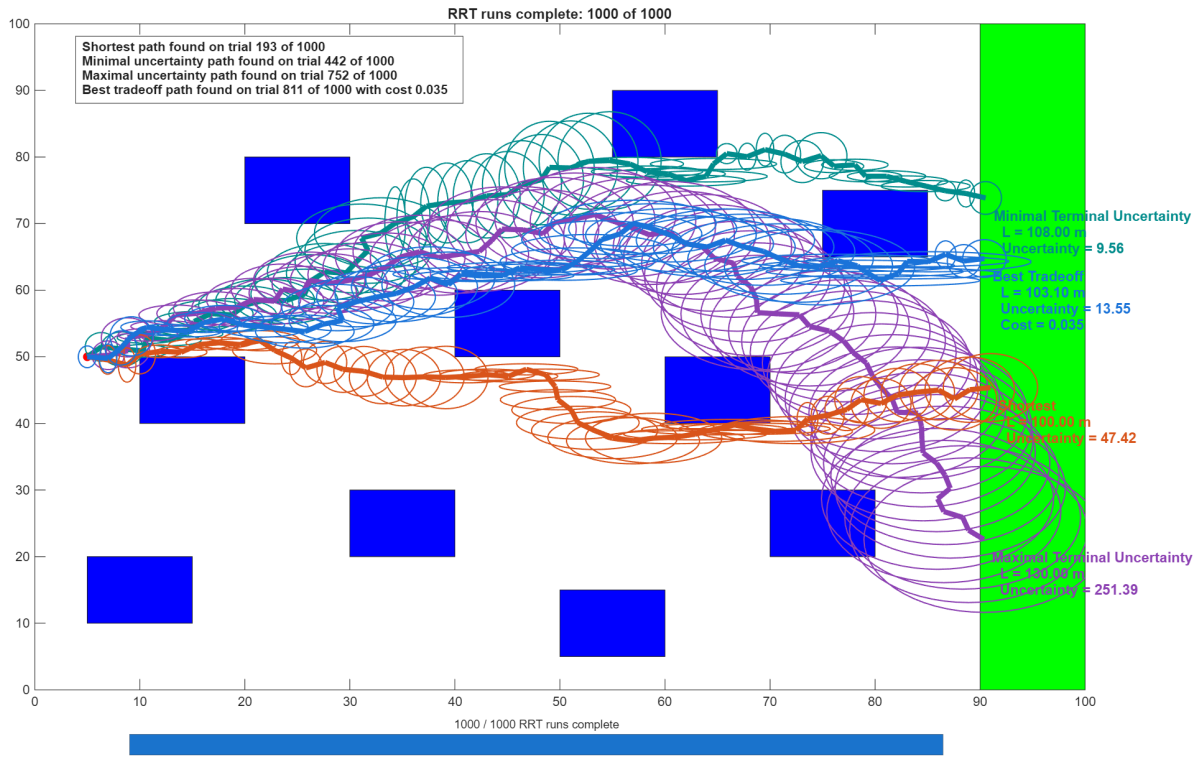

Figure 1: Selected RRT paths and Kalman filter covariance ellipses for the 1000-trial run.

The shortest path is the most direct route to the goal. It moves nearly horizontally across the workspace while avoiding obstacles along the way. This keeps the total distance low at 100 m, but the terminal uncertainty is higher than both the minimum-uncertainty and best-tradeoff paths. The shortest and minimum-uncertainty paths pass comparably near obstacles, but they do so at different points in the workspace.

The minimum-uncertainty path takes a longer route, with a path length of 108 m, but ends with much lower terminal uncertainty. A key difference is that near the end of its path, the robot observes an obstacle closer to the goal than it does on the shortest path. This gives the Kalman filter additional position information later in the trajectory, which reduces the final covariance.

The maximum-uncertainty path is both long and poorly localized. It appears to spend the longest duration outside useful sensing range, especially toward the end of the path. During these portions of the trajectory, process noise continues to increase the covariance while there are few or no position measurements to reduce it. As a result, this path reaches a terminal uncertainty of 251.39.

3 Distance, Uncertainty, and Collision Risk

Traveling near obstacles can improve localization because the robot can compare range-beam observations against the known map. When an obstacle is visible from a horizontal beam, the robot effectively gains information about its x-position. When an obstacle is visible from a vertical beam, it gains information about its y-position. These measurements reduce the corresponding components of the error covariance matrix.

This same behavior also introduces collision risk. Because the robot has nonzero localization uncertainty, the actual robot position may differ from the nominal planned state. If a covariance

ellipse intersects an obstacle, then there is a plausible localization error under which the robot could be in collision even though the nominal path is collision-free. Therefore, a good path should not simply stay as close as possible to obstacles. It should use obstacle visibility enough to reduce uncertainty while still maintaining reasonable clearance.

Additional distance is worthwhile only when it produces a meaningful reduction in uncertainty. A longer path that does not pass through informative sensing regions can increase distance without improving localization. To capture this tradeoff, the best-tradeoff path was selected using

$$J = w_d \bar{d} + w_u \bar{u},$$

where $\bar{d}$ is normalized path length, $\bar{u}$ is normalized terminal uncertainty, and $w_d = w_u = 0.5$ for this run. With equal weighting, the best-tradeoff path had a path length of 103.1 m and terminal uncertainty of 13.55. This is slightly longer than the shortest path, but it substantially reduces terminal uncertainty. It is also less certain than the minimum-uncertainty path, but it achieves most of the uncertainty reduction with less additional distance.

The most reasonable weighting depends on the application. A time-critical task would place more weight on distance, while a precision navigation task would place more weight on uncertainty. For this run, the equal-weighted best-tradeoff path is a useful middle ground between minimizing travel distance and minimizing terminal covariance.

4 Number of RRT Trials

Increasing the number of RRT trials improved the quality of the selected paths. With more trials, the algorithm had more opportunities to sample paths that were smoother, more direct, or more informative from a localization standpoint. In this workspace, 1000 trials produced noticeably better paths than 100 trials.

Based on these results, 100 trials is useful but not reliably sufficient for finding a high-quality path. More than 100 trials gives the selection process a better set of candidate paths, especially when optimizing for both path length and localization uncertainty. For this workspace, 1000 trials provided a more reliable basis for identifying the shortest, minimum-uncertainty, maximum-uncertainty, and best-tradeoff paths.

5 Assumptions

These observations are based on the SimLab 2 assumptions: the robot is modeled as a point, the workspace obstacles are known and rectangular, the robot follows the planned path exactly, and the Kalman filter is used to predict covariance growth rather than to correct a simulated perturbed trajectory. The results also depend on randomized RRT samples, so individual selected paths can change from run to run.

6 Included MATLAB Files

The MATLAB source associated with this report is organized as follows:

- `simlab2/simlab2.m`: driver script that runs the RRT trials, propagates covariance, selects representative paths, and plots results
- `simlab2/RRT.m`, `simlab2/Tree.m`, and `simlab2/TreeNode.m`: RRT implementation and tree data structures

- `simlab2/propagate_KF_path.m` and `simlab2/collect_sensor_data.m`: Kalman covariance propagation and obstacle-visibility measurement logic
- `simlab2/find_shortest_path.m` and `simlab2/find_best_path.m`: path-selection helpers
- `simlab2/plot_path.m`, `simlab2/plot_covariance_evolution.m`, and `simlab2/selection_color.m`: plotting utilities for selected paths and covariance ellipses
- `simlab2/generate_obstacles.m`, `simlab2/collision_check_point.m`, `simlab2/collision_check_segment.` and `simlab2/ellipse.m`: workspace, collision-checking, and ellipse helper functions

Disclosure of Assistance

Generative AI was used as a supplementary tool to help interpret the assignment prompt and assist with editing portions of this report. The MATLAB implementations, simulation runs, figure generation, and interpretation of the results were produced, verified, and submitted by me, Matthew Lepis.